

Generating Two Vectors with a Given Correlation

Goal

The goal is to construct two vectors whose sample Pearson correlation is exactly c , apart from numerical rounding.

The key idea is geometric:

$$r = \cos(\theta)$$

That is, after centering the vectors, the Pearson correlation is the cosine of the angle between them.

So we need to build two centered vectors with angle θ such that:

$$\cos(\theta) = c$$

Step 1: Generate x

```
x <- rnorm(n)
```

This creates a random vector.

We need one vector to serve as the first variable.

Step 2: Center x

```
x_centered <- x - mean(x)
```

Pearson correlation is not based on the raw vectors directly.

It is based on deviations from the mean:

$$x_i - \bar{x}$$

So centering is necessary because correlation ignores location.

After centering:

$$\sum_{i=1}^n x_i = 0$$

Step 3: Standardize the Length of x

```
x_standard <- x_centered / sqrt(sum(x_centered^2))
```

This does not standardize to variance 1 in the usual statistical sense.

It standardizes the vector length to 1:

$$\sum_{i=1}^n x_i^2 = 1$$

This makes later calculations simpler.

Since $x_standard$ has length 1, the projection formula becomes:

$$(z \cdot x)x$$

instead of:

$$\frac{z \cdot x}{x \cdot x} x$$

Step 4: Generate a Second Random Vector

```
z <- rnorm(n)
```

We need another direction in space to create variation in y that is not explained by x .

But we cannot use z directly, because it will usually have some accidental correlation with x .

Step 5: Center z

```
z_centered <- z - mean(z)
```

Again, correlation uses centered vectors.

Since `x_standard` is centered, z must also be centered before we control its relationship with x .

Step 6: Project z onto x

```
projection <- sum(z_centered * x_standard) * x_standard
```

This finds the part of $z_centered$ that points in the same direction as $x_standard$.

Geometrically:

$$z = \text{part parallel to } x + \text{part perpendicular to } x$$

The projection is the parallel part.

Step 7: Remove the Projection

```
z_orthogonal <- z_centered - projection
```

This removes from z everything that is correlated with x .

After this step:

$$x_{\text{standard}} \cdot z_{\text{orthogonal}} = 0$$

So $z_{\text{orthogonal}}$ is uncorrelated with x_{standard} .

This is essential because we want to build y from:

- one part aligned with x
- one part independent of x

Step 8: Standardize the Orthogonal Vector

```
z_standard <- z_orthogonal / sqrt(sum(z_orthogonal^2))
```

Now both vectors have length 1:

$$\|x_{\text{standard}}\| = 1$$

$$\|z_{\text{standard}}\| = 1$$

and they are orthogonal:

$$x_{\text{standard}} \cdot z_{\text{standard}} = 0$$

Step 9: Construct y

```
y <- c * x_standard + sqrt(1 - c^2) * z_standard
```

The vector y is built as a mixture of:

- x_{standard} , perfectly correlated with x
- z_{standard} , uncorrelated with x

The coefficient c controls how much of y points in the direction of x .

The coefficient

$$\sqrt{1 - c^2}$$

ensures that y also has length 1.

Why the Correlation Becomes Exactly c

The correlation between x_{standard} and y is:

$$r = x_{\text{standard}} \cdot y$$

Substitute the formula for y :

$$r = x_{\text{standard}} \cdot \left(cx_{\text{standard}} + \sqrt{1 - c^2} z_{\text{standard}} \right)$$

Distribute:

$$r = c(x_{\text{standard}} \cdot x_{\text{standard}}) + \sqrt{1 - c^2}(x_{\text{standard}} \cdot z_{\text{standard}})$$

Final Result

Since:

$$x_{\text{standard}} \cdot x_{\text{standard}} = 1$$

and:

$$x_{\text{standard}} \cdot z_{\text{standard}} = 0$$

we obtain:

$$r = c(1) + \sqrt{1 - c^2}(0)$$

Therefore:

$$r = c$$

The algorithm deliberately constructs y so that its geometric angle with x produces the desired Pearson correlation